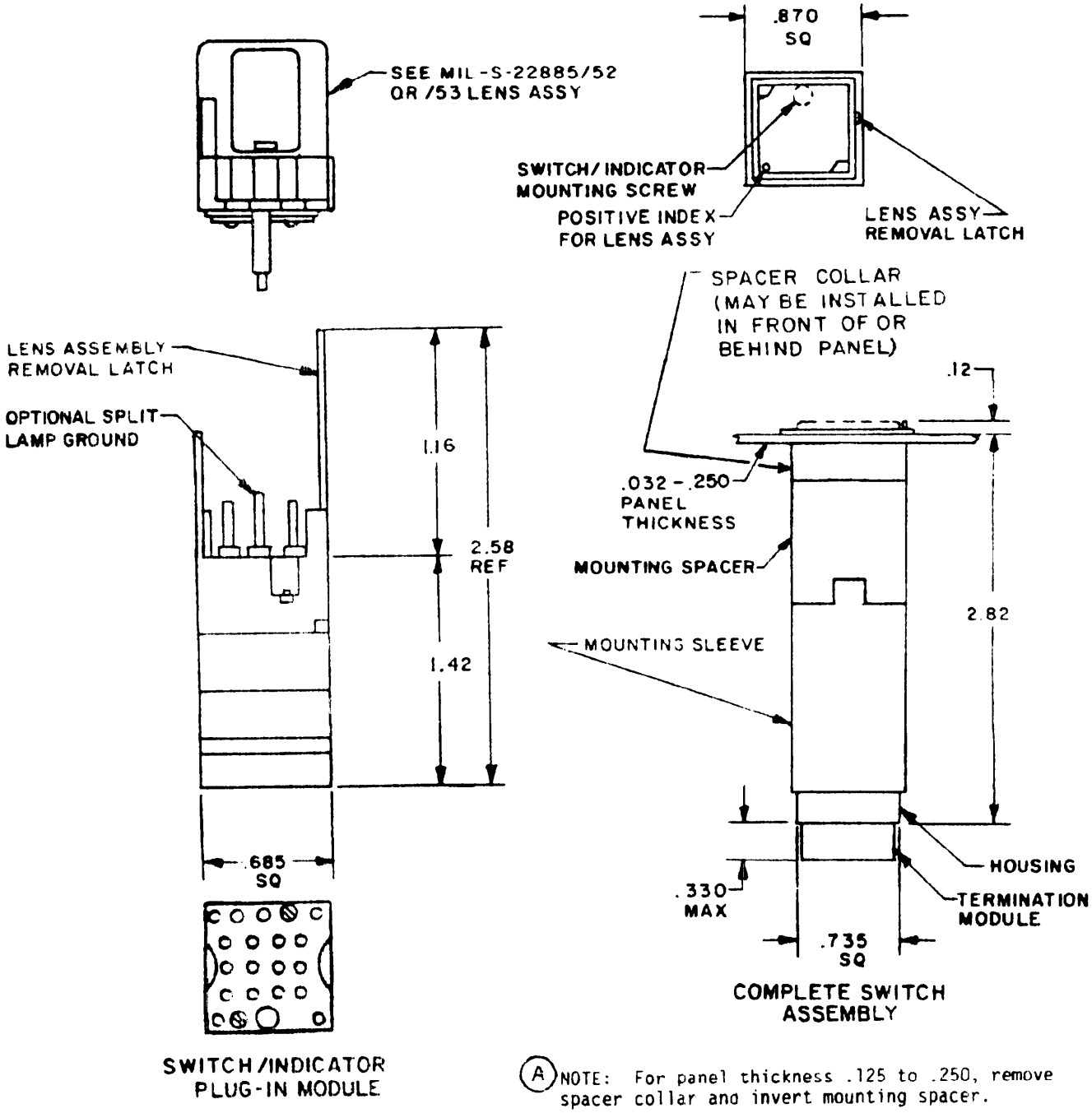
MILITARY SPECIFICATION SHEET

SWITCHES, PUSHBUTTON, ILLUMINATED, 2PDT AND 4PDT, PLUG-IN MODULE (COMMON TERMINATION SYSTEM)

This specification is approved for use by Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the switches described herein shall consist of this specification and the latest issue of MIL-S-22885.

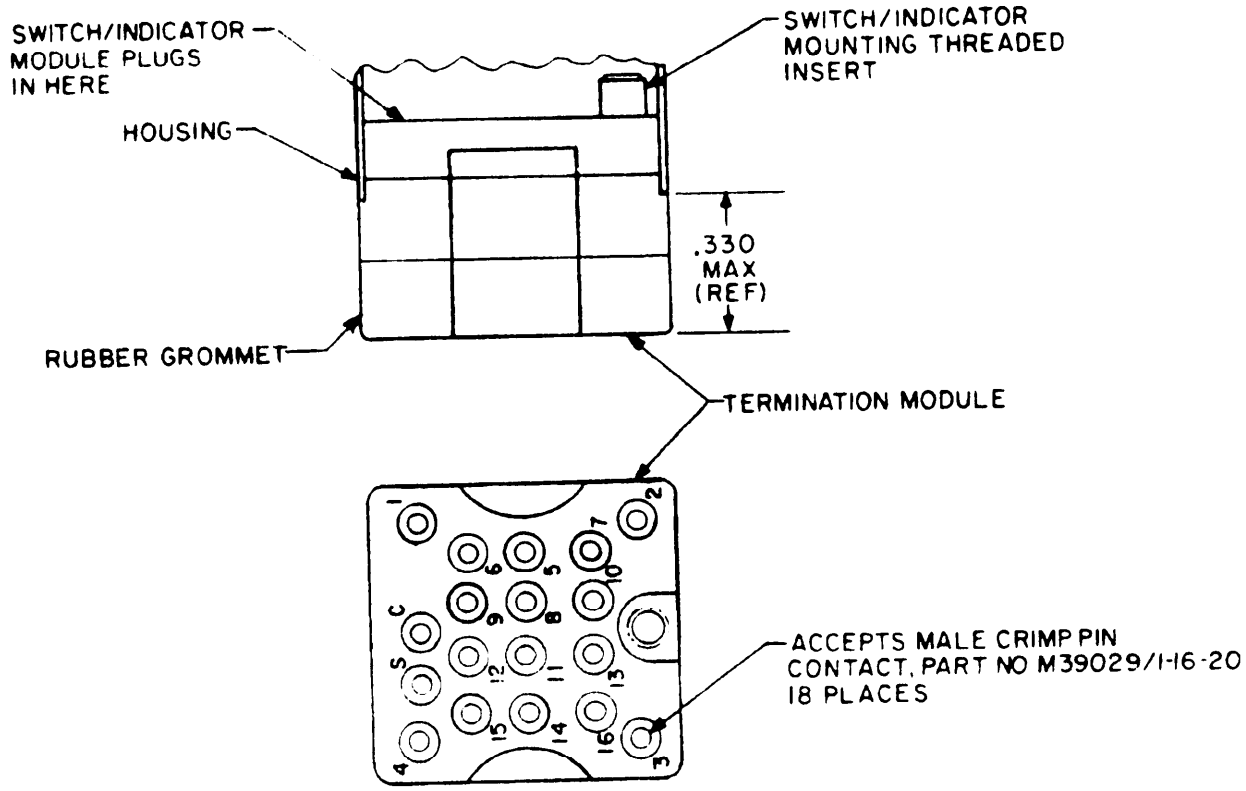


(A) NOTE: For panel thickness .125 to .250, remove spacer collar and invert mounting spacer.

FIGURE 1. Pushbutton switch assembly.

FSC 5930

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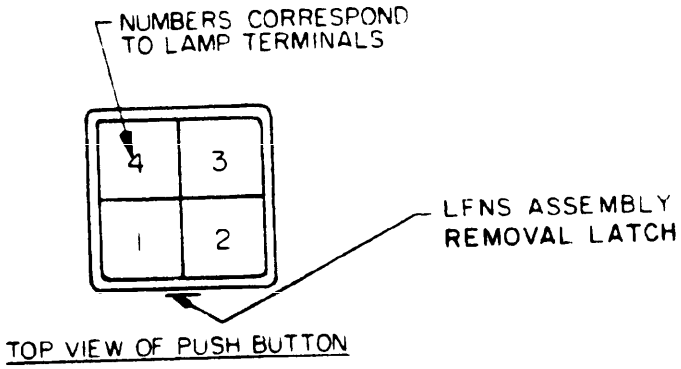
TERMINAL IDENTIFICATION

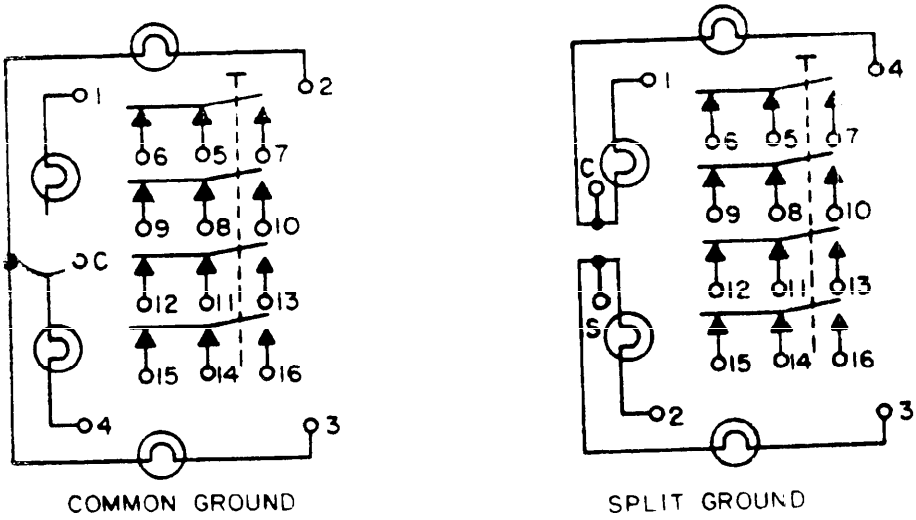
INCHES	MM
.032	.81
.12	3.0
.125	3.18
.250	6.35
.330	8.38
.685	17.40
.735	18.67
.870	22.10
1.16	29.5
1.42	36.1
2.58	65.5
2.82	71.6

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Unless otherwise specified, tolerances are ±.015 (.38 mm) for three place decimals and ±.03 (.8 mm) for two place decimals.
4. Terminals shall be permanently identified with the numbers shown.
5. The design configuration is optional within the envelope dimensions shown.

FIGURE 1. Pushbutton switch assembly - Continued.

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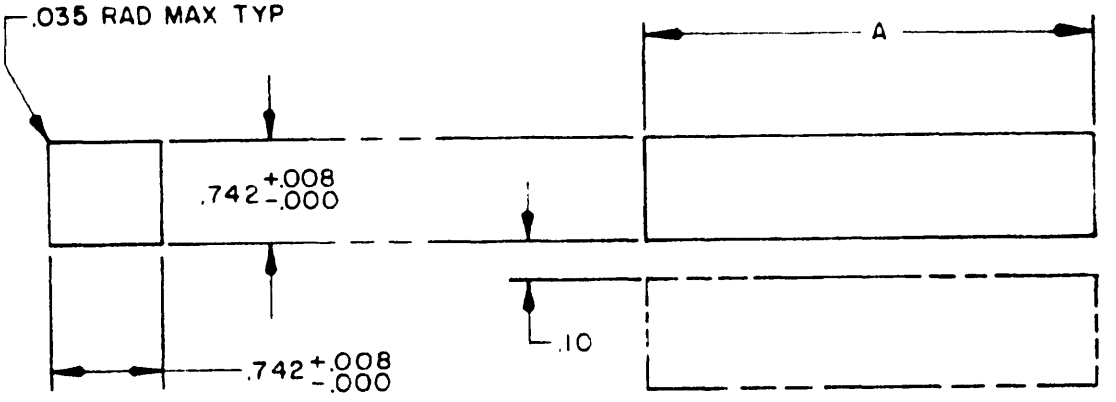




NOTES:
1. Terminals 8 thru 13 are omitted from 2PDT configurations.
2. Terminals 5 thru 16 are omitted for indicator configurations.
3. Circuit schematic shall be marked on the housing.
4. Terminals 5, 8, 11, and 14 are pole terminals.
5. Terminal S omitted on common ground circuit.

FIGURE 2. Lamp and 4PDT switch schematics.

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NO. UNITS	DIMENSION A
1	.742
2	1.612
3	2.482
4	3.352
5	4.222
6	5.092
n	A = .742 - .870 (n-1)

INCHES	MM
.008	.20
.035	.89
.10	2.5
.742	18.85
.870	22.10
1.612	40.94
2.482	63.04
3.352	85.14
4.222	107.24
5.092	129.34

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Unless otherwise specified, tolerances are ±.015 (.38 mm) for three place decimals and ±.03 (.8 mm) for two place decimals.
4. Dimensions apply for both vertical and horizontal (shown) slot mounting.

FIGURE 3. Horizontal and vertical slot mounting.

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DESIGN FEATURES:

Enclosure design: 1 (unsealed).

Temperature characteristic: 1 (-55°C to +85°C).

Vibration grade: 3 (10 - 2,000 Hz).

Lens transmittance: T (translucent).

Complete illuminated pushbutton switch assembly shall consist of:

1 Lens assembly in accordance with MIL-S-22885/52 or /53. Design permits application of photographic or engraved legends.

1 Switch/indicator module with mounting screw.

1 Housing with termination module, mounting sleeve, mounting spacer, and spacer collar.

4 Lamps (T-1-3/4 midget-flange base) incandescent. Not included, order separately.

REQUIREMENTS:

Material and finish:

Lens assembly: In accordance with MIL-S-22885/52 or /53.

Display screen: Clear transparent thermoplastic material.

Housing: Finished black.

Switch/indicator module housing and spacer collar:

Non-RFI shielded - Black anodized per MIL-A-8625.

RFI shielded - Chemical film coated per MIL-C-5541.

Mounting sleeve and mounting spacer: Corrosion-resistant steel.

Termination module: In accordance with MIL-STD-1549. (Termination module accepts male crimp pin contacts, part No. M39029/1-16-20.)

Lamp socket contacts - Copper alloy; nickel plated .0001 inch thick per QQ-N-290, class 2.

Switch/indicator socket contacts - Copper alloy; gold plated per MIL-G-45204, type II, grade C, class 1, over nickel plate .000050 inch thick.

Termination module grommet - Silicone rubber per ZZ-R-765.

Design and construction:

Dimensions and configuration: See figures 1, 2, and 3 and table I.

Weight: 0.120 pound ±10 percent (including lens assembly and four lamps).

Electrical ratings (at sea level):

Contact rated load	28 Vdc	115 Vac
Resistive	2.0 amperes	2.0 amperes
Inductive	1.0 ampere	1.0 ampere
Lamp	0.75 ampere	0.75 ampere

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Performance:

Operating characteristics:

Actuation force: 3.5 ±1.0 pounds.

Actuator travel: 0.200 inch maximum.

Thermal shock: During high temperature portion of test, all four lamps in lens assembly shall be energized with maximum rated voltage.

Shock: Method I.

Dielectric withstanding voltage: 450 Vrms at 70,000 feet.

RFI shielding (when applicable, see table I):

Requirement: When parts are tested in accordance with the following, the attenuation shall be not less than specified on figure 4.

Test method: Parts shall be tested to determine shielding effectiveness in accordance with the general procedures specified in MIL-STD-285. The test shall consist of electric field and plane wave measurements performed at the following frequencies:

Electric field	Minimum attenuation	Plane wave	Minimum attenuation
15 kHz	60 dB	100 MHz	47 dB
150 kHz	60 dB	400 MHz	39 dB
500 kHz	60 dB	1 GHz	33 dB
1 MHz	60 dB	2.5 GHz	28 dB
10 MHz	60 dB	10 GHz	20 dB

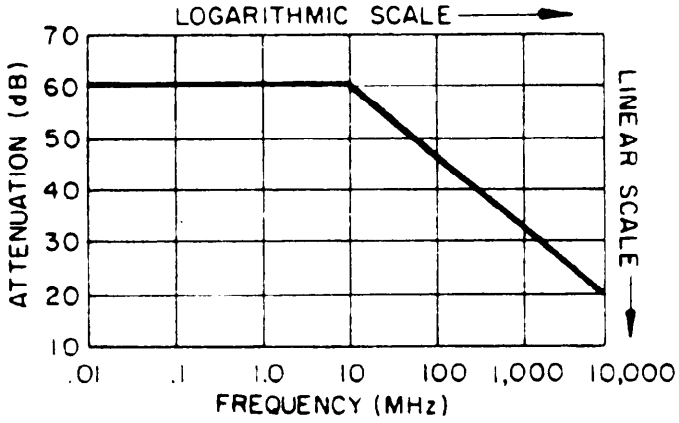


FIGURE 4. Shielding effectiveness.

Marking:

Part number: M22885/95-(dash number from table I). The complete part number for a switch module with color filters is shown in the following example:

EXAMPLE: M22885/95-034 R G G Y

4PDT, alternate action, non-RFI shielded, common ground in lamp circuit, 4-way split lens

Red filter in position 1

Green filter in position 2

Green filter in position 3

Yellow filter in position 4

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TABLE I. Dash numbers and design and operating characteristics.

Dash number		Circuit description	Lens code (see table II)	RFI shielded
mmon ground	Split ground			
WITHOUT LENS ASSEMBLIES				
-001	-201	4PDT momentary	---	---
-002	-202	4PDT alternate	---	---
-003 1/	-203 1/	2PDT momentary	---	---
-004	-204	2PDT alternate	---	---
-005	-205	Indicator	---	---
-006	-098	Housing	---	No
-007 2/	-099 2/	Housing	---	Yes
-008	-100	4PDT momentary	---	No
-009	-101	4PDT alternate	---	No
-010	-102	2PDT momentary	---	No
-011	-103	2PDT alternate	---	No
-012	-104	Indicator	---	No
-013	-105	4PDT momentary	---	Yes
-014	-106	4PDT alternate	---	Yes
-015	-107	2PDT momentary	---	Yes
-016	-108	2PDT alternate	---	Yes
-017	-109	Indicator	---	Yes
WITH LENS ASSEMBLIES 3/				
-018(XXXX)	-110(XXXX	4PDT momentary	A	No
-019(XXX)	-111(XXX)	4PDT momentary	B	No
-020(XX)	-112(XX)	4PDT momentary	C	No
-021(X)	-113(X)	4PDT momentary	D	No
-022(XXX)	-114(XXX)	4PDT momentary	E	No
(XX)	-115(XX)	4PDT momentary	F	No
(XXX)	-116(XXX)	4PDT momentary	G	No
-025(XXX)	-117(XXX)	4PDT momentary	H	No
-026(XXXX)	-118(XXXX)	4PDT momentary	A	Yes
-027(XXX)	-119(XXX)	4PDT momentary	B	Yes
-028(XX)	-120(XX)	4PDT momentary	C	Yes
-029(X)	-121(X)	4PDT momentary	D	Yes
-030(XXX)	-122(XXX)	4PDT momentary	E	Yes
-031(XX)	-123(XX)	4PDT momentary	F	Yes
-032(XXX)	-124(XXX)	4PDT momentary	G	Yes
-033(XXX)	-125(XXX)	4PDT momentary	H	Yes
-034(XXXX)	-126(XXXX)	4PDT alternate	A	No
-035(XXX)	-127(XXX)	4PDT alternate	B	No
-036(XX)	-128(XX)	4PDT alternate	C	No
-037(X)	-129(X)	4PDT alternate	D	No
-038(XXX)	-130(XXX)	4PDT alternate	E	No
-039(XX)	-131(XX)	4PDT alternate	F	No
-040(XXX)	-132(XXX)	4PDT alternate	G	No
-041(XXX)	-133(XXX)	4PDT alternate	H	No
-042(XXXX)	-134(XXXX)	4PDT alternate	A	Yes
-043(XXX)	-135(XXX)	4PDT alternate	B	Yes
-044(XX)	-136(XX)	4PDT alternate	C	Yes
-045(X)	-137(X)	4PDT alternate	D	Yes
-046(XXX)	-138(XXX)	4PDT alternate	E	Yes
-047(XX)	-139(XX)	4PDT alternate	F	Yes
-048(XXX)	-140(XXX)	4PDT alternate	G	Yes
-049(XXX)	-141(XXX)	4PDT alternate	H	Yes
-050(XXXX)	-142(XXXX)	2PDT momentary	A	No
-051(XXX)	-143(XXX)	2PDT momentary	B	No

ootnotes at end of table.

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TABLE I. Dash numbers and design and operating characteristics - Continued.

Dash number		Circuit description	Lens code (see table II)	RFI shielded
Common ground	Split ground			
-052(XX)	-144(XX)	2PDT momentary	C	No
-053(X)	-145(X)	2PDT momentary	D	No
-054(XXX)	-146(XXX)	2PDT momentary	E	No
-055(XX)	-147(XX)	2PDT momentary	F	No
-056(XXX)	-148(XXX)	2PDT momentary	G	No
-057(XXX)	-149(XXX)	2PDT momentary	H	No
-058(XXXX)	-150(XXXX)	2PDT momentary	A	Yes
-059(XXX)	-151(XXX)	2PDT momentary	B	Yes
-060(XX)	-152(XX)	2PDT momentary	C	Yes
-061(X)	-153(X)	2PDT momentary	D	Yes
-062(XXX)	-154(XXX)	2PDT momentary	E	Yes
-063(XX)	-155(XX)	2PDT momentary	F	Yes
-064(XXX)	-156(XXX)	2PDT momentary	G	Yes
-065(XXX)	-157(XXX)	2PDT momentary	H	Yes
-066(XXXX)	-158(XXXX)	2PDT alternate	A	No
-067(XXX)	-159(XXX)	2PDT alternate	B	No
-068(XX)	-160(XX)	2PDT alternate	C	No
-069(X)	-161(X)	2PDT alternate	D	No
-070(XXX)	-162(XXX)	2PDT alternate	E	No
-071(XX)	-163(XX)	2PDT alternate	F	No
-072(XXX)	-164(XXX)	2PDT alternate	G	No
-073(XXX)	-165(XXX)	2PDT alternate	H	No
-074(XXXX)	-166(XXXX)	2PDT alternate	A	Yes
-075(XXX)	-167(XXX)	2PDT alternate	B	Yes
-076(XX)	-168(XX)	2PDT alternate	C	Yes
-077(X)	-169(X)	2PDT alternate	D	Yes
-078(XXX)	-170(XXX)	2PDT alternate	E	Yes
-079(XX)	-171(XX)	2PDT alternate	F	Yes
-080(XXX)	-172(XXX)	2PDT alternate	G	Yes
-081(XXX)	-173(XXX)	2PDT alternate	H	Yes
-082(XXXX)	-174(XXXX)	Indicator	A	No
-083(XXX)	-175(XXX)	Indicator	B	No
-084(XXX)	-176(XXX)	Indicator	C	No
-085(X)	-177(X)	Indicator	D	No
-086(XXX)	-178(XXX)	Indicator	E	No
-087(XX)	-179(XX)	Indicator	F	No
-088(XXX)	-180(XXX)	Indicator	G	No
-089(XXX)	-181(XXX)	Indicator	H	No
-090(XXXX)	-182(XXXX)	Indicator	A	Yes
-091(XXX)	-183(XXX)	Indicator	B	Yes
-092(XX)	-184(XX)	Indicator	C	Yes
-093(X)	-185(X)	Indicator	D	Yes
-094(XXX)	-186(XXX)	Indicator	E	Yes
-095(XX)	-187(XX)	Indicator	F	Yes
-096(XXX)	-188(XXX)	Indicator	G	Yes
-097(XXX)	-189(XXX)	Indicator	H	Yes

1/ These plug-in switch/indicator modules do not include the housings and lens assemblies.

2/ These housings do not include the switch/indicator modules and lens assemblies.

3/ These dash numbers are for switch/indicator modules with color filters. The X (X's) shall be replaced with the desired color symbol(s) of MIL-S-22885 in the position sequence shown in table II.

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TABLE II. Lens code, style, and color position sequence.

LENS CODE	LENS STYLE	COLOR SYMBOL POSITION 1/
A	4 - WAY SPLIT	
B	3 - WAY SPLIT	
C	SPLIT HORIZONTALLY	
D	SOLID	
E	3 - WAY SPLIT	
F	SPLIT VERTICALLY	
G	3 - WAY SPLIT	
H	3 - WAY SPLIT	

1/ Lens assembly removal latch is shown for orientation.

LIFICATION:

Qu ification inspection: All applicants for qualification approval shall (istrate that each of their items conforms to all the requirements sp ified in the applicable documents, singularly and in combination with all other previously qualified items, regardless of manufacturer.

Group submission: Table III applies when the momentary and alternate action switches and the indicator are submitted for qualification at the same time.

TABLE III. Qualification inspection.

Part number	Qualification inspection table X of MIL-S-22885		Extent of approval
	Group	Number of sample units	
85/95-037(X) PDT alternate, non-RFI ielded, common ground)	I	22	All
	II	4 (from group I)	
	III	2 (from group I)	
	IV	2 (from group I)	
	V	2 (from group I)	
	VI	12 (from group I)	
	VIII	10 1/	
85/95-121(X) PDT momentary, RFI ielded, split ground)	I	10	
	II	4 (from group I)	
	III	2 (from group I)	
	VII	2 (from group I)	
	2/	2 (from group I)	

ample units which have been subjected to other tests may be used.

ample units shall be subjected to the RFI shielding test of this specification he

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QUALITY CONFORMANCE:

Group C inspection: Group C inspection shall be in accordance with MIL-S-22885 and table IV herein.

TABLE IV. Group C inspection.

Part number and number of sample units	Tests of MIL-S-22885, table XII, as identified by sample numbers	Extent of approval
M22885/95-037(X) 10 sample units (4PDT alternate, non-RFI shielded, common ground)	Sample numbers 1 through 10	All
M22885/95-121(X) 4 sample units (4PDT momentary, RFI shielded, split ground)	Sample numbers 1, 2, 9, and 10 1/	

1/ Two sample units shall be subjected to the RFI shielding test of this specification sheet.

NOTES:

1. For high shock applications, see MIL-S-24317/10.
2. Revision letters are not used to denote changes due to the extensivenss of the changes.

Custodian:
Navy - EC

Preparing activity:
Navy - EC

Review activities:
Navy - AS
DLA - ES

(Project 5930-N547)

Agent:
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